```
#Create a Python program to extract metadata
#(size, DPI, format) from an image file using
Pillow.


from PIL import Image

img = Image.open("photo.png")

print("Image format:", img.format)
print("Image size:", img.size)

dpi_value = img.info.get("dpi")

if dpi_value:
    print("Image DPI:", dpi_value)
else:
    print("DPI information not available")
```

```
#Create a list of sample items (e.g., KYC document
names)
#and use a for loop to print each item.
documents = ["Aadhaar Card", "PAN Card", "Voter
ID",
             "Passport", "Driving License"]

for doc in documents:
    print("Required document:", doc)
```

```
#Given a folder containing 5-10 sample KYC images,
use a
#for loop to run OCR on each file and store
#the results in a list.


from PIL import Image

img = Image.open("sample.png")

print("Image format:", img.format)
print("Image size:", img.size)

dpi_value = img.info.get("dpi")

if dpi_value:
    print("Image DPI:", dpi_value)
else:
    print("DPI info not found!!")
```

```
#Write a Python script to load and display an
image
#using OpenCV (cv2.imread, cv2.imshow).

import cv2

image = cv2.imread("photo.png")

if image is None:
    print("Cannot open image")
else:
    cv2.imshow("Image", image)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

```
#Write a small calculator to add, subtract,
multiply,
#and divide two numbers input by the user.


a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)

if b != 0:
    print("Division:", a / b)
else:
    print("Division not possible (cannot divide by
zero)")
```

```
#3. Store user information in a dictionary (name,
age, ID)
#and print each key-value pair.

user = {}

user["name"] = input("Enter name: ")
user["age"] = input("Enter age: ")
user["id"] = input("Enter ID: ")

for key in user:
    print(key, ":", user[key])
```

AI-Powered Identity Verification and Fraud Detection for KYC Compliance

Internship Project – Group 2

What is the main concept?

- Verify whether a person’s identity is real or fake.
- Detect fraud in documents like AADHAR , PAN, Passport , etc,…
- Help banks , trading platforms, and digital payment apps follow KYC rules properly.
- KYC makes sure a company knows who their customer really is.
- Finding fake addresses, manipulated documents, suspicious patterns

Problem Statement

- Fake identity documents are increasing.
- We come across AADHAR address details to be common.
- Even the scanners in the AADHAR card were also scammed and people essential details were lost.
- Manual verification is slow and costly and verifying all the individuals is time consuming
- Some fields like banks , government related sectors need faster and more accurate verification systems.

Purpose of the project

- Automate the identity verification process in KYC systems.
- Reduce the usage of manual work and replace with fast and secure AI.
- Reduce the chance of fraud detection of misled of human essential information.
- Securing the case sensitive data in all the real world scenarios.
- Reduce human errors in identity verification.
- Create a scalable solution that can handle large volumes of KYC applications.

AI concepts to be used

- Natural Language Processing (NLP)- Extract address from AADHAR details
  -Check if address format is valid
- Computer Vision (CV)- Support OCR for text extraction
  -Process scanned AADHAR cards
- Graph Neural Networks (GNN)- Find hidden fraud patterns
  - Detect unusual connections between customers
- Optical Character Recognition (OCR)-Extract name, address, DOB from AADHAR
  -Convert scanned documents into usable data
- Machine Learning / Deep Learning - Training fraud detection models
  - Improving detection accuracy over time
- Azure OpenAI Integration- Text validation
  - Integrate with AML/KYC systems

Working Process

- Once the system has clean and structured information like name, address, DOB, and ID number, it can automatically verify whether the details are consistent, valid, and genuine.
- It helps in detecting mismatches, duplicate identities, fake addresses, and suspicious patterns. This reduces the chances of identity fraud and prevents unauthorized individuals from opening accounts.
- Beyond just verification, KYC also helps in risk assessment and compliance monitoring. The extracted data can be used to check AML rules, monitor suspicious activities, and generate alerts if something unusual is detected.
- It supports faster onboarding, improves security, and ensures the organization follows regulatory requirements properly. So after data extraction, KYC plays a major role in both fraud prevention and maintaining legal compliance.

Real Research Publications and academic papers

- AI-Driven Fraud Detection in e-KYC with Integrated Fingerprint
- Fraud Prevention at Scale: AI/ML Integration in Customer Identity Verification
- Digital Identity Verification Using Machine Learning to Reduce Fraud in Micro-Lending
- Deepfake Detection in KYC Application
- AI Applications for e-KYC and Identity Fraud Detection – Literature Review
- AI-Based Identity Fraud Detection: A Systematic Review

Difference with research publications and this proposed system

The research papers you saw mainly focus on **one specific area** of identity verification. For example:

Papers focus only on **biometric verification** (face, fingerprint, voice), **deepfake detection**, general fraud detection using **ML models,** literature reviews explaining existing techniques.

Many of them **discuss ideas theoretically but don't describe full system deployment.**

- This Proposed system focuses on:
  - NLP for document text analysis
  - Computer Vision for document image verification
  - GNN for detecting relationship-based fraud
  - Integration into real AML/KYC systems
  - AADHAR address verification
  - Real-time onboarding integration

CONCLUSION….

- In this project, we worked on improving the traditional KYC process using AI. By using technologies like NLP, Computer Vision, and Graph Neural Networks, we aimed to automate identity verification and detect fraudulent address details more effectively.
- The system reduces manual effort, speeds up customer onboarding, and improves accuracy in fraud detection. Overall, this project shows how AI can be practically applied to make KYC processes more secure, efficient, and reliable for real-world use in every sector.